

Data Science Analysis Report: Sentiment, Risk, and Trading Behavior

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1. Objective

The objective of this analysis is to understand how market sentiment (Fear vs. Greed) relates to trade behavior, risk-taking, and profitability. By merging a historical trade dataset with a fear–greed index, we aim to identify behavioral patterns and derive actionable trading signals that can inform risk management and position sizing.

2. Dataset Overview

The analysis utilizes two primary datasets. The first is a historical trade log containing detailed information for each executed trade, including timestamps, instrument, trade size in tokens and USD, side (long/short), execution price, and realized profit or loss (closedPnL). The second dataset is a daily fear–greed index, providing a sentiment classification (e.g., Fear or Greed) and a corresponding numeric value per calendar date. After preprocessing, the datasets are merged on a common date field, yielding a unified table of trades where each record is annotated with the prevailing sentiment on the trade date.

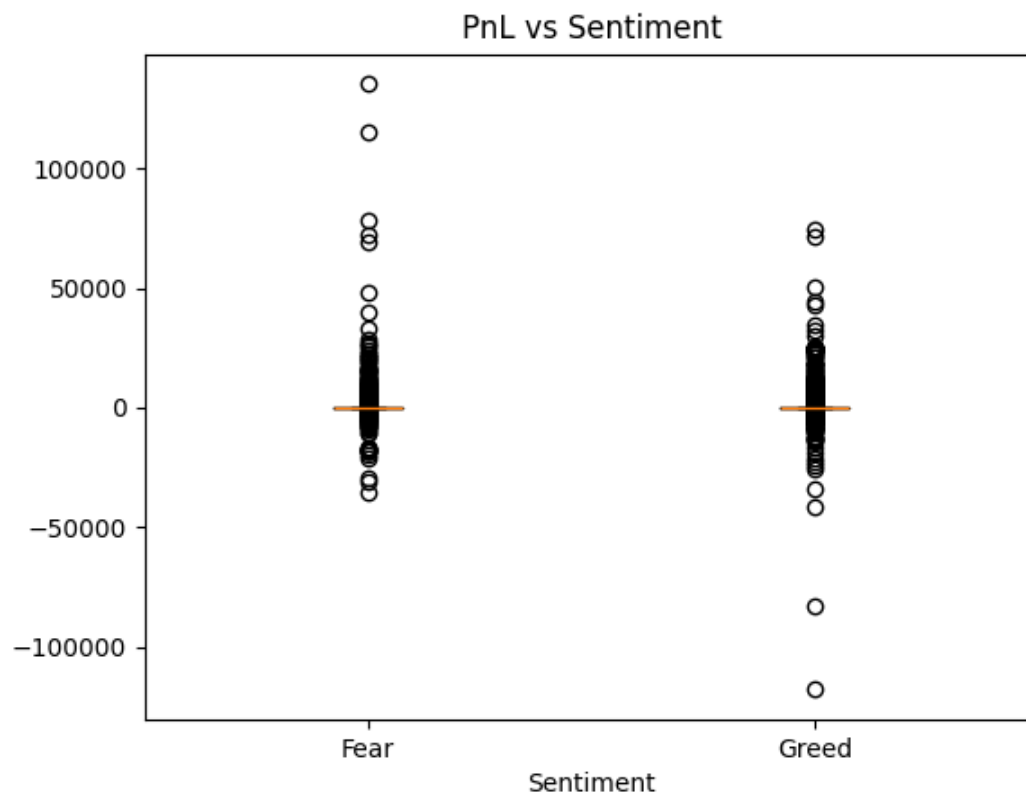
3. Methodology

The workflow begins with loading the raw CSV files into Pandas DataFrames, followed by standardizing column names and data types for consistency. Timestamp fields are converted to proper datetime objects, and calendar dates are extracted to serve as the join key across datasets. An inner join on date is then performed to retain only those trades for which a corresponding sentiment observation exists. Subsequent steps include deriving additional features that capture trade volume, risk exposure, and directional profitability, and computing summary statistics grouped by sentiment to reveal differences between Fear and Greed regimes. Visualizations are generated to compare distributions and aggregated metrics across these regimes.

4. Feature Engineering

Several engineered variables are created to better represent trading behavior. Trade volume is defined as the absolute value of the position size ($\text{trade_volume} = |\text{size}|$), providing a scale-invariant measure of trade magnitude. A binary profit_flag is introduced to indicate whether a trade is profitable ($\text{closedPnL} > 0$). A risk_score proxy is defined as the absolute USD notional of each trade ($\text{risk_score} = |\text{size_usd}|$), capturing the monetary exposure per position. The qualitative sentiment classification is encoded numerically (Fear $\rightarrow 0$, Greed $\rightarrow 1$) to facilitate group-wise aggregation and modeling. These features form the foundation for subsequent statistical analysis and visualization.

Visualization 1: PnL Distribution by Sentiment

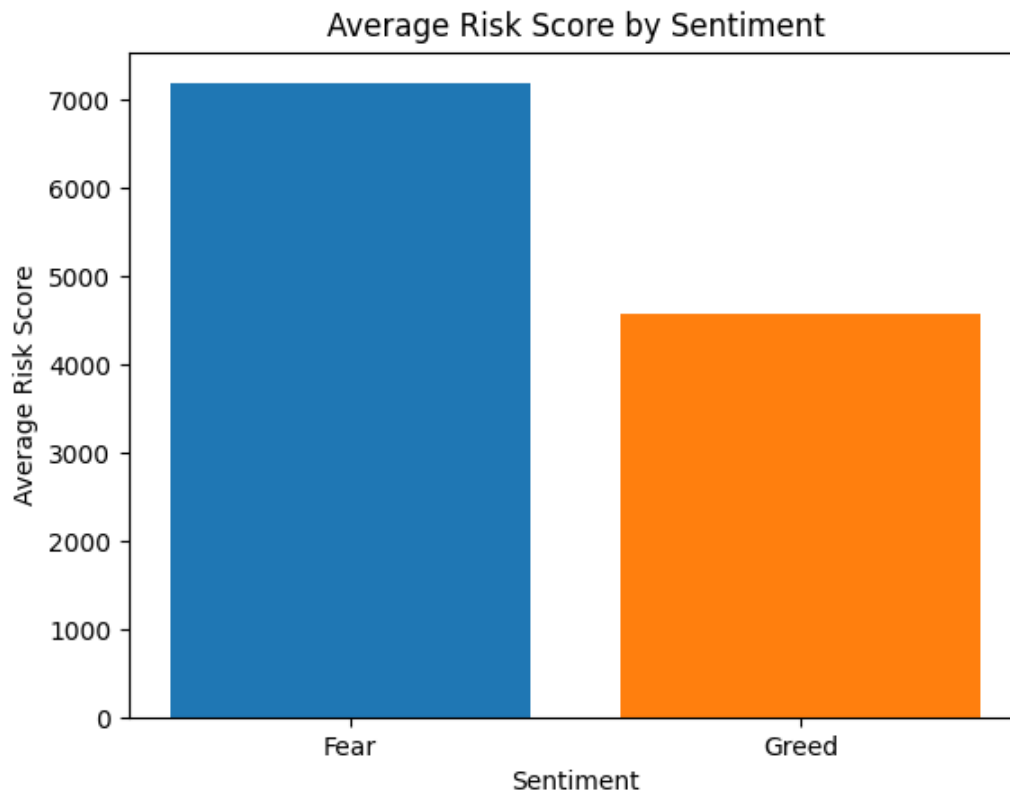


5. Key Observations

Aggregated statistics reveal meaningful differences in performance and behavior across sentiment regimes. Mean and median closedPnL, computed by sentiment, highlight whether periods labeled as Greed are associated with systematically higher or lower profitability compared with Fear periods. Average and total trade_volume by sentiment indicate whether traders tend to scale up or down their positions depending on market mood. The variance of trade_volume by sentiment serves as a proxy for volatility in trade sizing, suggesting whether position sizing becomes more erratic in euphoric or fearful conditions. In addition, the joint distribution of risk_score and closedPnL allows identification of

high-exposure trades that result in losses, which are especially informative for risk management.

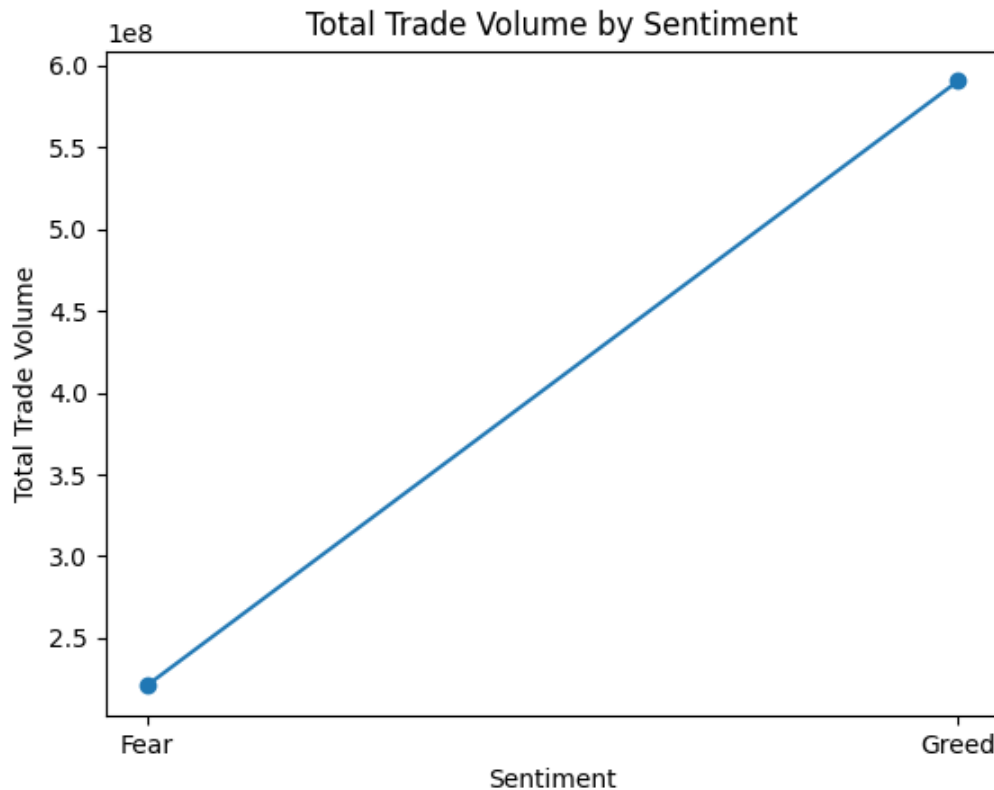
Visualization 2: Average Risk Score by Sentiment



6. Behavioral Insights

The joint analysis of sentiment, trade volume, and realized PnL suggests that trader behavior is influenced by prevailing market mood. In Greed regimes, traders may exhibit a tendency toward larger position sizes and higher aggregate risk_score, reflecting increased risk appetite. If profitability does not scale proportionally with this additional risk, it implies overconfidence and potential mispricing of downside risk. In Fear regimes, reduced average trade_volume and risk_score may indicate defensive positioning and risk aversion, possibly at the cost of underutilized capital or missed opportunities. The distribution of losing trades across sentiment states provides additional evidence on whether poor outcomes cluster disproportionately in euphoric conditions or during risk-off periods.

Visualization 3: Total Trade Volume by Sentiment



7. Actionable Trading Signals

Based on the engineered features and grouped statistics, several practical signals can be derived. First, dynamic position sizing rules can be calibrated so that maximum `risk_score` per trade is capped more tightly during Greed regimes, mitigating the impact of overleveraging in euphoric markets. Second, systematic de-leveraging triggers can be implemented when sentiment transitions from Greed to Fear while recent closed PnL has been negative, signaling vulnerability to drawdowns. Third, monitoring the ratio of profitable to unprofitable trades (`profit_flag` distribution) by sentiment can serve as an early warning indicator: a deterioration in this ratio under Greed conditions may suggest a late-stage cycle where risk-taking is no longer compensated.

8. Limitations

This analysis is subject to several limitations. The fear–greed index is a coarse proxy for collective sentiment and may not fully capture intraday shifts or instrument-specific dynamics. The merged dataset includes only those dates for which both trade data and sentiment data are available, potentially introducing selection bias. The `risk_score` metric is a simplified measure based solely on nominal exposure and does not incorporate volatility, correlation, or portfolio context. Furthermore, the analysis is observational; any inferred relationships between sentiment and performance are correlational rather than causal and may be influenced by unobserved confounders such as regime changes, liquidity conditions, or strategy modifications.

9. Conclusion

By integrating a sentiment indicator with detailed trade-level data, this project provides a structured view of how market mood aligns with risk-taking and realized outcomes. The engineered features and grouped statistics reveal that sentiment regimes can meaningfully influence trade volume, risk exposure, and the distribution of profits and losses. While the results must be interpreted with caution given the limitations, they support the design of sentiment-aware risk controls and position sizing frameworks. Future work could extend this analysis by incorporating additional market variables, testing predictive models of PnL conditioned on sentiment, and validating the robustness of the derived trading signals across different instruments and time periods.